

Repo Rate and Money supply - A Study On Indian Prospective

Jitendra Kumar Dixit,¹ Dr.Sanjeev Gupta ²

¹ Assistant Professor, GLA University, Mathura

² Professor ,MLB Govt. College of Excellence, Gwalior(M.P.)

E mail: jitendra.dxt@gmail.com,

Contact No-08171549777

Abstract

The Central Bank is endeavor to maintain smooth functioning of the economy seesaws between resigning inflation and money supply. Globalization & integration of economics make the task more complex. Recession, collapse of economies & soaring inflation make the situation more bitter. Central bank use different policy rates (repo rate, reverse repo rate, CRR, SLR) to maintain the money supply within the economy.

Through the study, an effort has been made to analyze the relation between repo rate and money supply (M2) and the reaction of repo rate variation on various component of money supply (M2). For the analysis, sample data covers a period from first quarter of financial year 2000-01 to third quarter of financial year 2013-14. Repo rate and database on Indian economy is collected from the web portals of Reserve Bank of India (RBI). For the analysis, unit root test followed by co- integration is used to examine the long term relationship between repo rate and monetary aggregates. The vector auto regressive (VAR) model and Granger causality test is to investigate the dynamic inter relationship between repo rate and money supply. The result suggests the repo rate variation provides a platform for controlling the money supply with the economy.

Key word – liquidity, money supply, Granger causality,

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Jitendra Kumar Dixit,¹ Dr. Sanjeev Gupta ²

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1-Introduction

Policy rates are the key tool in the hands of economist for managing the balance among different economics activities. Among them, repo rate is effective and efficient tool in implementing monetary policy with in the economy. With some economic activities repo rate is directly associated with some other in-directly, but the equilibrium among the economic activities can be achieved through repo rate variation.

Moneyiness in the economy can also be maintained via repo rate variation, as repo rate is associated with different monetary aggregates. As the monetarist of 1960s and 70s, Friedman and Schwartz (1963), and Cagan (1972), emphasized monetary aggregates as indicators of monetary policy. Estrella and Mishkin (1996) suggested that the fundamental role of money is : as information variables, as indicator of policy actions and as instruments in a policy rule. Feldstein and Stock (1994) used M2 as component of money supply in analyzing the relation with GDP. In the continuum, M2 is used as variable depicting money supply.

1.1 Policy Rate (Repo Rate)

Repo is defined as an agreement in which one party sells securities or other assets to a counterparty, and simultaneously commits to repurchase the same asset, at an agreed future date at a repurchase price. The said repurchase price would cover the original sell price plus a return on the use of the sale proceeds during the term of the repo.

Repos can be divided into four broad categories -

(a) Classic Repo (US style); (b) Buy-Sell Back Repo (Indian market follows this type) and (c) Securities Lending for a fee and (d) Tri-party Repo.

Classic repo involves an initial sale of securities with a simultaneous agreement to repurchase them at a later date where the start and end prices of the securities are the same and a separate

payment of "interest" is made. The principal difference between a repurchase agreement and a buy/sell- back stem from the fact that repurchase agreements are always documented, while buy/sell-backs are not required to be documented as there are implicitly two separate contracts. Under a Tripartite repo, a common custodian /clearing agency arranges for custody as well as clearing and settlement of repos transactions.

1.2- Money Supply (M2)

Narrow Money (M1) = Currency with the Public + Demand Deposits with the Banking System + 'Other' Deposits with the RBI.

$M2 = M1 + \text{Savings Deposits of Post-office Savings Banks.}$

2.Purpose of the Study- The purpose of the study is to explore the relation between repo rate and money supply (M2) .

3.Research Methodology

3.1. Sampling Design- Sampling Design for the present study was as follows:

- a) The study encompasses the money supply (M2) and policy rate (repo rate).
- b) Selection of sample for money supply (M2) and policy rate (repo rate) has been judgment sampling.

3.2. Statistical Design- The statistical design used for the study is as follows:

- a) For money supply (M2) and their association with repo rate, secondary data was required which was collected from Database on Indian Economy provided by Reserve Bank of India (RBI).
- b) Quarterly data was composed for duration ,first quarter of financial year 2000-01 to third quarter of financial year 2013-14 (2000-01(Q1)-20013-14(Q3)).
- c) Statistical measures like Augmented Dickey-Fuller Test ,VAR & Granger Causality Tests have been used through E-Views7.

4 . Methodology and Analysis-

This study investigates the association between money supply (M2) and repo rate for India during the period ,first quarter of financial year 2000-01 to third quarter of financial year 2013-14 (2000-01(Q1)-20013-14(Q3)).

To check the stationary properties (unit roots) of time series Augmented Dickey- Fuller (ADF) test Dickey and Fuller, 1979) is employed. It consists of running a regression of the first difference of the series .This can be expressed as:

$$\Delta y_t = \alpha_1 y_{t-1} + \sum_{j=1}^p \beta_j \Delta y_{t-j} + \varepsilon_t \quad \dots\dots\dots (1)$$

The test for unit root is conducted on the coefficient of (y_{t-1}) in the above equation (eqⁿ-1). If the coefficient is different from zero then the hypothesis of a unit root is rejected. Rejection of the null hypothesis reflects stationary.

Now, the time series has to be analyzed for co-integration to identify the long-run relationships between the variables and to avoid any scope of spurious regression. In this stage, the Johansen (1988) co-integration test is used to identify a co-integrating relationship among the variables.

According to Granger (1969), if a variable Y help to forecast another variable X, then we say that Y Granger-causes X. If Y causes X and X does not causes Y, it is said unidirectional causality. If Y does not cause X and X does not cause Y, then both the variable are statistically independent. If Y causes X and X causes Y, it is said that feedback exists between the variables.

For Granger test, a autoregressive lag length q (or p) is assumed and OLS estimated following model-

$$X_t = \lambda_1 + \sum_{i=1}^q a_{1i} X_{t-i} + \sum_{j=1}^q b_{1j} Y_{t-j} + \mu_{1t} \quad \dots\dots\dots (2)$$

$$Y_t = \lambda_2 + \sum_{i=1}^q a_{2i} X_{t-i} + \sum_{j=1}^q b_{2j} Y_{t-j} + \mu_{2t} \quad \dots\dots\dots (3)$$

Hassapis et al. (1999) show that in the absence of co-integration, the direction of causality can be decided upon via standard F-tests in the first differenced VAR. The VAR in the first difference can be written as:

$$\left\{ \begin{array}{l} \Delta X_t = \lambda_1 + \sum_{i=1}^k a_{1i} \Delta X_{t-i} + \sum_{j=1}^k b_{1j} \Delta Y_{t-j} + \mu_{1t} \quad \dots\dots\dots (4) \end{array} \right.$$

$$\left\{ \begin{array}{l} \Delta Y_t = \lambda_2 + \sum_{i=1}^p a_{2i} \Delta X_{t-i} + \sum_{j=1}^p b_{2j} \Delta Y_{t-j} + \mu_{2t} \quad \dots\dots\dots (5) \end{array} \right.$$

5. Empirical Results-

In regression analysis of time series data, stationarity of a time series is minimum requirement to minimize the problem of spurious result, that might arise if the series is non-stationary. Hence, time series should be first made stationary. Stationarity of time series refers, mean and variance should be constant and co-variance depends on the distance of two time periods.

The Augmented Dicky-Fuller unit root test (Dickey and Fuller, 1979) is used to check stationarity of the time series and this is required for any time series for VAR analysis. Repo rate and M2 is stationary at 1st difference at 5% significance level. So, repo rate and M2 both are integrated of order one i.e I(1).

For selecting optimal lag, traditionally VAR model is repeated by reducing the lag length value from large value to minimum. We select the optimal lag for which Akaike information criterion and Schwarz criterion have smallest value (Grasa, 1989; DeJong et al., 1992; Maddala and Gujarati, 2003). In this study optimal lag is selected based on five criteria; Sequential modified LR test statistics, Final prediction error (FPE), Akaike information criterion (AIC), Schwarz criterion (SC) and Hannan-Quinn information criterion (HQ) that is suggested by Lutkepohl (1993). For the present study selected optimal lag is 1.

Table no-1.1 & 1.2 reflect the output of Johansen co-integration test on Repo rate and M2. Both maximum eigen value and trace statistics suggest that no co-integration exist between both the variables at 5% significance level.

Table-1.1 Unrestricted Co integration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.**
None	0.197126	14.69783	15.49471	0.0657
At most 1 *	0.095127	4.598205	3.841466	0.0320

Trace test indicates no cointegration at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Table-1.2 Unrestricted Co-integration Rank Test (Maximum Eigen value)

Hypothesized No. of CE(s)	Eigenvalue	Max-Eigen Statistic	0.05 Critical Value	Prob.**
None	0.197126	10.09963	14.26460	0.2055
At most 1 *	0.095127	4.598205	3.841466	0.0320

Max-eigenvalue test indicates no cointegration at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

As the output of Johansen co-integration suggests, there is no co-integration between both the variable ,so VAR model is most suitable to identify the association between both the variables, repo rate and M2. Model generate two set of equations, in which only one equation depict some meaningful interpretation. That equation is as follows-

$$M2 = C(4)*REPO(-1) + C(5)*M2(-1) + C(6) \dots\dots\dots(6)$$

For identifying the bilateral or unidirectional causality among the repo rate and M2, Granger causality (1987) is applied. The result is table no-2 suggest that there is a causality association between repo rate and M2. Repo rate can be use-full in forecasting money supply within the economy and variation in repo rate affect the supply of money.

Table-2 Pair-wise Granger Causality Tests

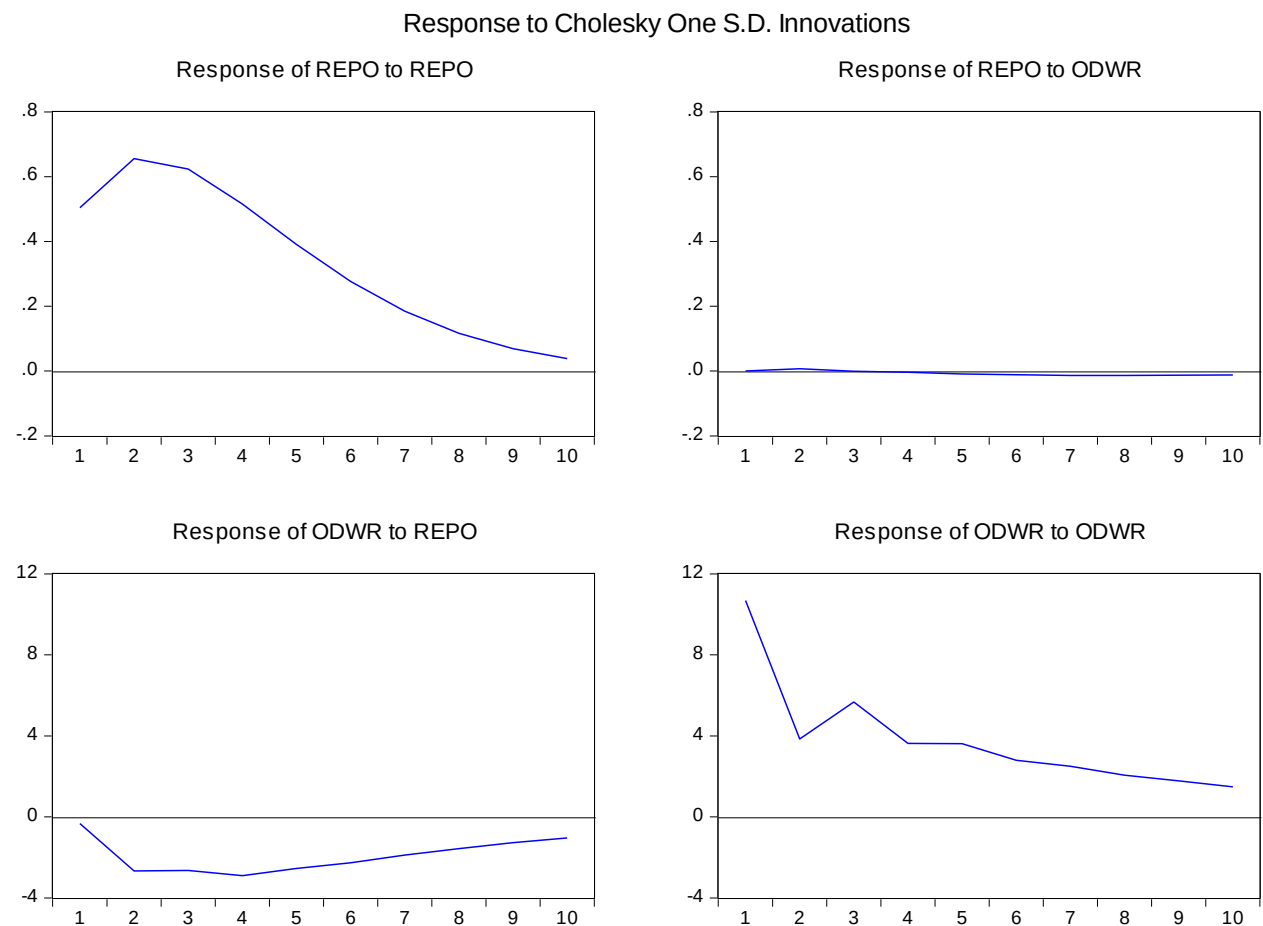
Null Hypothesis:	F-Statistic	Prob.
M2 does not Granger Cause REPO	1.96331	0.1682
REPO does not Granger Cause M2	4.75090	0.0347

Impulse response function shows the response of one standard deviation shock in the error term of the other variable. The x-axis shows the time period in quarters and y-axis shock in the movement trend. In fig.-1,the movement of repo rate to its own shock causes positive fluctuations. Shock in repo rate causes negative fluctuation in other deposits with RBI. The effect is realized after second quarter and after that fluctuation in ODWR remains negative until another shock. Shock in ODWR causes negative fluctuation in repo rate. The movement of ODWR to its own shock causes positive fluctuation that keep on declining quarter after quarter.

In fig.-2, the shock in repo rate causes positive fluctuation on currency with public that is realized with some time lag. Positive fluctuation in CWP is seems to be negative long run. Shock in CWP causes negative fluctuation up to second quarter and become positive after that. The movement of CWP to its own shock causes positive fluctuation.

In fig.-3, the movement of demand deposit to its own shock and repo rate causes positive fluctuations. Shock in demand deposits causes positive fluctuation in repo rate that is realize after first quarter. Shock in repo rate causes negative fluctuation in demand deposits that is realized with time lag. The movement of repo rate to its own shock causes positive fluctuations.

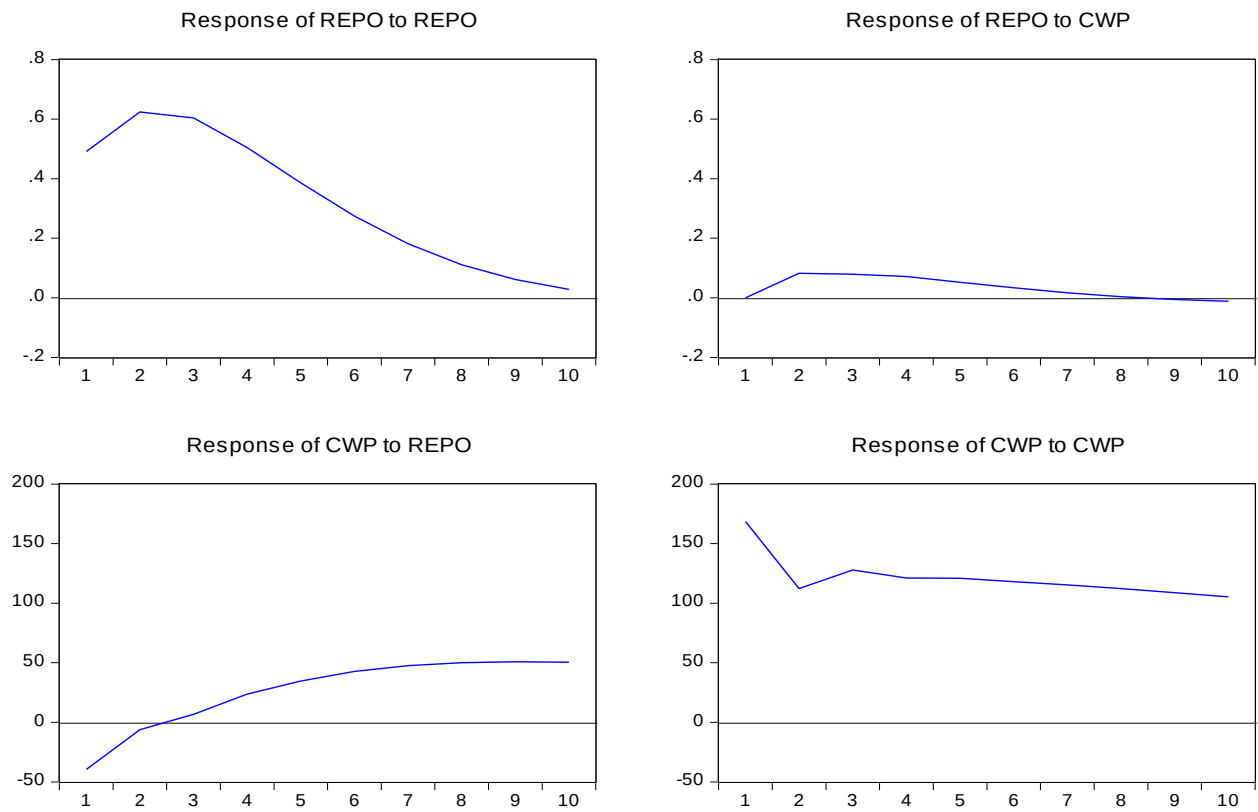
In fig.-4, shock in repo rate causes positive fluctuation in money supply that is realized after first quarter and become negative in long run .Shock in M2 causes negative fluctuation in repo that becomes positive after second quarter. The movement of M2 to its own shock causes positive fluctuation.



OWDR- other deposits with RBI

Fig. no -1

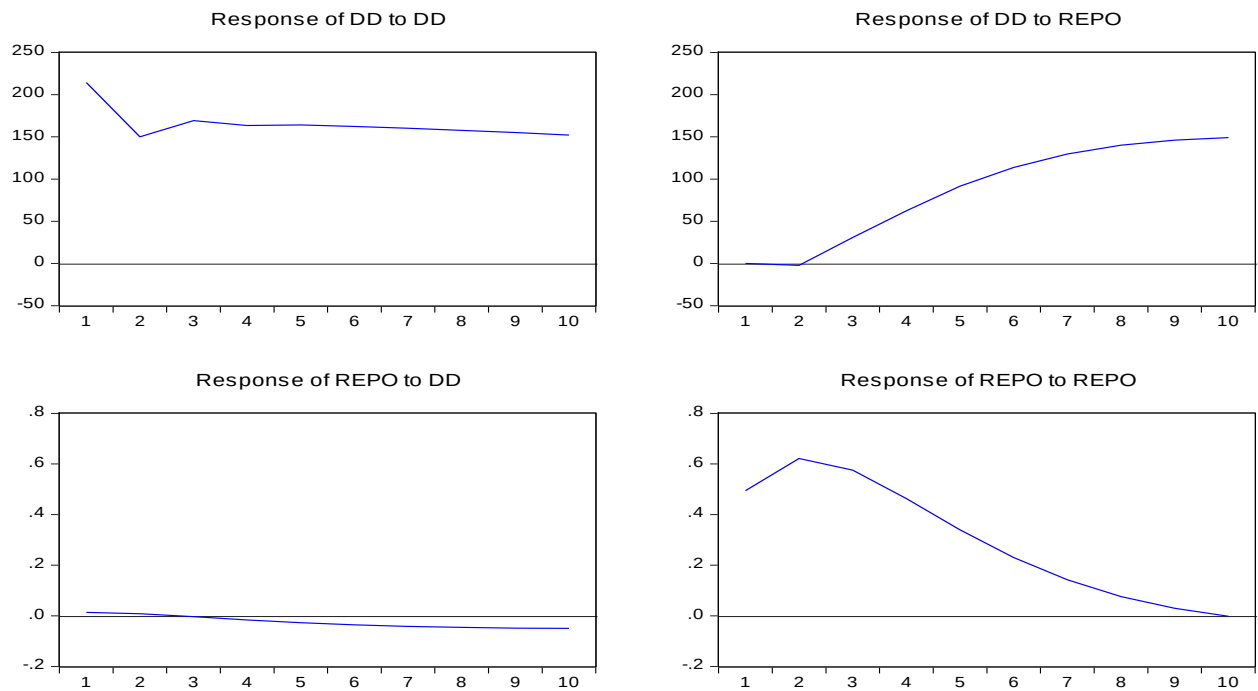
Response to Cholesky One S.D. Innovations



CWP= currency with public

Fig. no--2

Response to Cholesky One S.D. Innovations



DD= demand deposits

Fig.-3

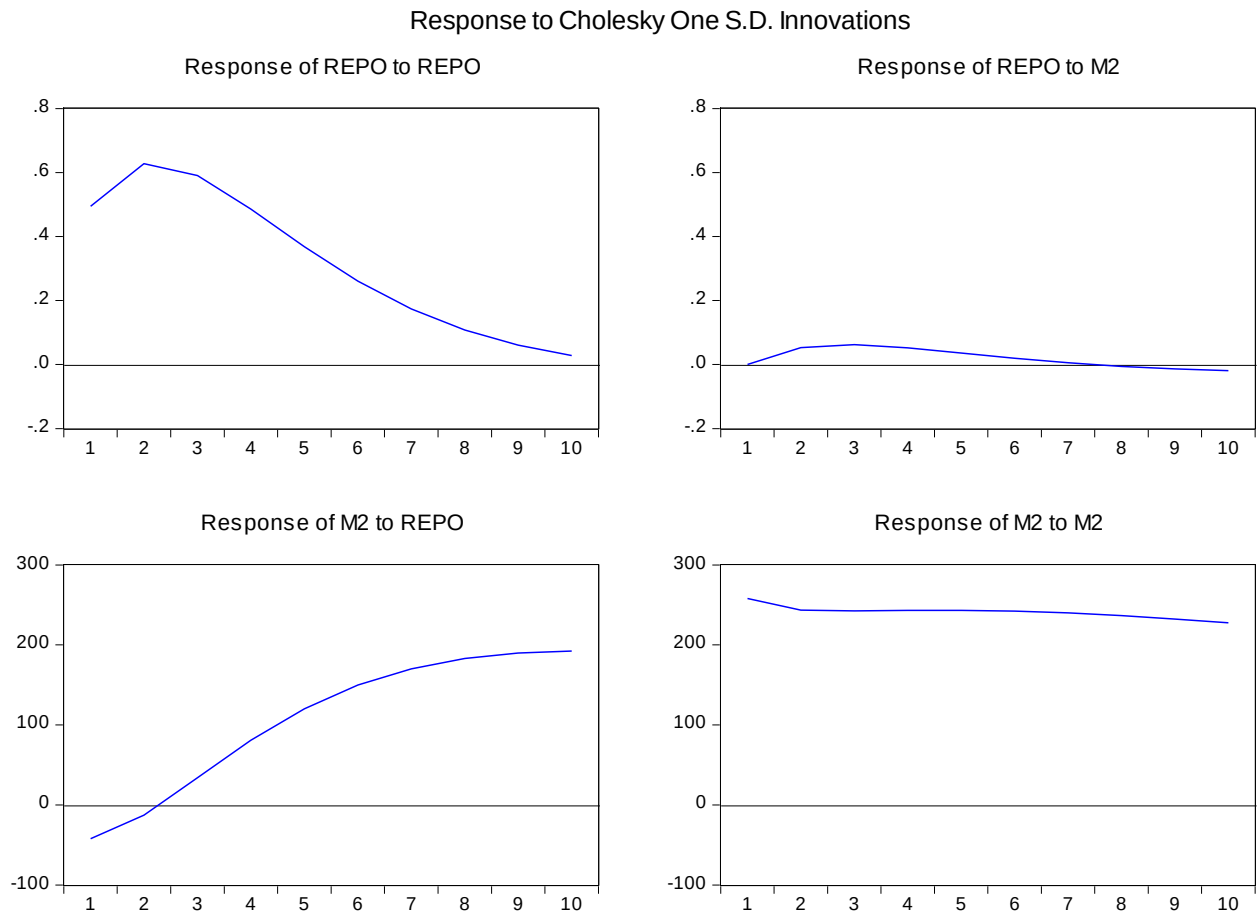


Fig. no-4 Impulse Response Function

6. Conclusion –

Variation in repo rate directly affects demand deposits, currency with the public and other deposits with RBI via interest rate channel. Components which are weakly correlated with repo rate controlled via some other route as repo rate is providing platform for setting other policy rates. The study suggests that the role of repo rate cannot be overlooked as it is strongly associated with other policy rates. Repo rate is base rate for controlling money supply (M2) with in the economy.

References-

- Dickey, D.A. and W.A. Fuller (1979). Distribution of the Estimators for Autoregressive Time Series with a Unit Root, *Journal of the American Statistical Association*, 74, pp. 427-431.
- Canova, F. (1995). Vector Autoregressive Models: Specification, Estimation, Inference, and Forecasting, in M.H. Pesaran and M. Wickens (eds.), *Handbook of Applied Econometrics: Macroeconomics*, Blackwell, Oxford UK & Cambridge USA.
- Engle, R.F. and Granger, C.W.J., 1987, 'Co-integration and Error-Correction: Representation, Estimation and Testing', *Econometrica*, 55, pp. 251-256.
- Jadhava N. (1988), "Definition and Measurement of Money Supply", *Economic and Political Weekly* 23(39) 2024, September 24
- Johansen, S., 1988, 'Statistical Analysis of Co-integration Vectors', *Journal of Economic Dynamics and Control*, 2, pp. 231-254.
- Report of the Working Group (Chairman Y. V. Reddy) on "*Money Supply: Analytics and Methodology of Compilation*", Reserve Bank of India, Mumbai, June 1998.
- Reserve Bank of India (1961), "Report of The First Working Group", *Reserve Bank Of India Bulletin* 15 1045-1066 and 1214-1219, July and August
- Reserve Bank of India (1977), *Report of the Second Working Group*, Bombay
- Reserve Bank of India (2008), *Handbook of Monetary Statistics of India*, Mumbai
- Srivastava N. N. (1978), "A Note on the Second Working Group Report on Money Supply in India: Concepts, Compilation and Analysis" in Economics Department ed. (1978), *Recent Developments in Monetary Policy and Theory*, Reserve Bank of India, Bombay, pp 175-184
- Shrestha B., (2010) *Impact of Money Supply on GDP and Price: Case of Nepal*. M. Phil. Nepal: Tribhuvan University.
- Time Series with a Unit Root', *Journal of the American Statistical Association*, 74, pp. 427-431
- Vasudevan A. (1980), 'Money Stock and Its Components in India, 1950-51 to 1979-80: A Statistical Account', *The Indian Economic Journal* 27(1) 1-13, July

APPENDIX A

Table A1- Augmented Dickey-Fuller Unit Root Test Results

Null Hypothesis: D(REPO) has a unit root

Exogenous: Constant

Lag Length: 0 (Automatic - based on SIC, maxlag=9)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-4.513519	0.0007
Test critical values: 1% level	-3.581152	
5% level	-2.926622	
10% level	-2.601424	

*MacKinnon (1996) one-sided p-values.

Table A2- Augmented Dickey-Fuller Unit Root Test Results

Null Hypothesis: D(M2) has a unit root

Exogenous: Constant

Lag Length: 0 (Automatic - based on SIC, maxlag=9)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-5.903329	0.0000
Test critical values: 1% level	-3.581152	
5% level	-2.926622	
10% level	-2.601424	

*MacKinnon (1996) one-sided p-values.

Table A3 Coefficient Table

Dependent Variable: M2

Method: Least Squares

	Coefficient	Std. Error	t-Statistic	Prob.
C(4)	71.51923	32.81216	2.179656	0.0347
C(5)	0.980725	0.008080	121.3696	0.0000
C(6)	-614.9400	251.0266	-2.449701	0.0183
R-squared	0.997026	Mean dependent var		9824.640
Adjusted R-squared	0.996890	S.D. dependent var		4597.863
S.E. of regression	256.3907	Akaike info criterion		13.99298
Sum squared resid	2892392.	Schwarz criterion		14.11108
Log likelihood	-325.8351	Hannan-Quinn criter.		14.03742
F-statistic	7374.656	Durbin-Watson stat		2.026911
Prob(F-statistic)	0.000000			

